

Showmik Bhowmik

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EDUCATION

Rochester Institute of Technology <i>Bachelor of Science in Computer Science. GPA: 3.78/4.0</i>	Rochester, N.Y. <i>Aug. 2022 – Dec. 2026</i>
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WORK EXPERIENCE

Software Engineer <i>Vira AI</i> - Built an AI video creative pipeline (Next.js, TypeScript, React, PostgreSQL/Prisma, AWS) that turns a client's photo and voice sample into a finished promotional video by chaining generative video (FAL/Kling), voice cloning (ElevenLabs), and lip sync into one automated workflow. - Developed webhook-driven async job orchestration coordinating long-running third-party AI jobs with per-job token validation, stage tracking, and reconciliation for stalled jobs, so generations survive client disconnects. - Worked with Terraform-provisioned AWS infrastructure — S3 with presigned direct browser uploads, IAM, Lambda video preprocessing, CloudFront — plus Stripe billing and Google OAuth authentication.	May 2026 – July 2026 <i>Seattle, WA (Remote)</i>
Software Engineer Co-op <i>LAPP Insulators</i> - Optimized backend algorithms, reducing API response time by 45%. - Improved scalability, query speed, and reliability for 10K+ daily users by migrating legacy SQL services to a distributed cloud-based database. - Refactored the in-house event-processing system, reducing latency by 25%; consolidated multiple services into a unified architecture, improving maintainability.	May 2025 – Aug. 2025 <i>LeRoy, N.Y.</i>
Software Engineer Co-op <i>Worldwide Electric</i> - Developed custom solutions using the Epicor/Prophet 21 framework and C#, integrating new user-facing functionality such as pick ticket logging. - Improved analytics by integrating custom logging with SQL Server Reporting Services (SSRS). - Led cross-functional meetings between multidisciplinary teams to align project scope and timelines.	May 2024 – Aug. 2024 <i>Rochester, N.Y.</i>
Student Lab Instructor <i>RIT Golisano College of Computing and Information Sciences</i> Mentored 25 students on assignment quality, debugging practices, and step-by-step code analysis; graded and gave detailed feedback for approximately 50 students on strict deadlines.	Aug. 2023 – Aug. 2025 <i>Rochester, N.Y.</i>

PROJECTS

Transcendent <i>RAG, Pinecone, Gemini 2.5 Flash, KMeans, A/B Testing</i> - Built a retrieval-augmented generation (RAG) system over Hacker News content using Pinecone vector search and Gemini 2.5 Flash, with KMeans clustering for topic grouping of retrieved results. - Ran A/B tests across retrieval and prompting configurations to measure answer-quality improvements.
VisionVoice <i>BLIP, SpeechT5, Gemini, Python, FastAPI, Angular</i> - Developed VisionVoice using AI models (Salesforce's BLIP, Microsoft's SpeechT5, Google's Gemini) to convert videos into accessible audio/text for users with visual impairments. - Built a low-latency delivery system with Python, FastAPI, and Angular; on a team of three, blended product management with technical solutions to drive the project's commercial introduction.
Gesture-Jutsu <i>MediaPipe, OpenCV, Random Forest, Real-Time ML</i> - Built a real-time hand gesture recognition system using MediaPipe and OpenCV that detects anime-inspired hand signs and triggers on-screen visual effects with 97% model accuracy. - Designed a full ML pipeline from scratch: custom data collection, Random Forest classifier training, and real-time inference processing 63 hand landmark coordinates per frame for low-latency predictions.

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C#, Go, TypeScript, JavaScript, SQL (Postgres), HTML/CSS
Web & Cloud: React, Next.js, Node.js, FastAPI, Angular, AWS (S3, Lambda, IAM, CloudFront), Terraform, Prisma, Stripe, OAuth
Machine Learning & AI: PyTorch, TensorFlow, JAX, LLM APIs, RAG, Pinecone, computer vision, NLP, speech processing
Developer Tools: Git, Docker, Azure DevOps, Maven, IntelliJ, VS Code, Visual Studio, Jupyter Notebook

HONORS AND AWARDS

- CS Honors Program - RIT - Dean's List (Fall 2022, Spring 2023, Fall 2024) - RIT - International Founders Scholarship Recipient - RIT - International Student Outstanding Service Award - RIT - Alpha Phi Omega Scholarship - RIT
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